Name: _____

Answer Sheet for AP® Chemistry Practice Exam #2, Section I

Question No.	Answer
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Question No.	Answer
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AP® Chemistry Exam

SECTION I: Multiple Choice

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

Instructions

Section I of this exam contains 60 multiple-choice questions. Pages containing a periodic table and lists containing equations and constants are also printed in this booklet.

Indicate all of your answers to the multiple-choice questions on the answer sheet. No credit will be given for anything written in this exam booklet, but you may use the booklet for notes or scratch work.

Use your time effectively, working as quickly as you can without losing accuracy. Do not spend too much time on any one question. Go on to other questions and come back to the ones you have not answered if you have time. It is not expected that everyone will know the answers to all of the multiple-choice questions.

Your total score on Section I is based only on the number of questions answered correctly. Points are not deducted for incorrect answers or unanswered questions.

At a Glance

Total Time
1 hour and 30 minutes
Number of Questions
60
Percent of Total Score
50%
Writing Instrument
Either pencil or pen with black or dark blue ink
Electronic Device
Calculator allowed

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DO NOT DETACH FROM BOOK.

PERIODIC TABLE OF THE ELEMENTS

1												13	14	15	16	17	18
1 **H** 1.008	2																2 **He** 4.00
3 **Li** 6.94	4 **Be** 9.01											5 **B** 10.81	6 **C** 12.01	7 **N** 14.01	8 **O** 16.00	9 **F** 19.00	10 **Ne** 20.18
11 **Na** 22.99	12 **Mg** 24.30	3	4	5	6	7	8	9	10	11	12	13 **Al** 26.98	14 **Si** 28.09	15 **P** 30.97	16 **S** 32.06	17 **Cl** 35.45	18 **Ar** 39.95
19 **K** 39.10	20 **Ca** 40.08	21 **Sc** 44.96	22 **Ti** 47.87	23 **V** 50.94	24 **Cr** 52.00	25 **Mn** 54.94	26 **Fe** 55.85	27 **Co** 58.93	28 **Ni** 58.69	29 **Cu** 63.55	30 **Zn** 65.38	31 **Ga** 69.72	32 **Ge** 72.63	33 **As** 74.92	34 **Se** 78.97	35 **Br** 79.90	36 **Kr** 83.80
37 **Rb** 85.47	38 **Sr** 87.62	39 **Y** 88.91	40 **Zr** 91.22	41 **Nb** 92.91	42 **Mo** 95.95	43 **Tc**	44 **Ru** 101.07	45 **Rh** 102.91	46 **Pd** 106.42	47 **Ag** 107.87	48 **Cd** 112.41	49 **In** 114.82	50 **Sn** 118.71	51 **Sb** 121.76	52 **Te** 127.60	53 **I** 126.90	54 **Xe** 131.29
55 **Cs** 132.91	56 **Ba** 137.33	57–71 *	72 **Hf** 178.49	73 **Ta** 180.95	74 **W** 183.84	75 **Re** 186.21	76 **Os** 190.23	77 **Ir** 192.22	78 **Pt** 195.08	79 **Au** 196.97	80 **Hg** 200.59	81 **Tl** 204.38	82 **Pb** 207.2	83 **Bi** 208.98	84 **Po**	85 **At**	86 **Rn**
87 **Fr**	88 **Ra**	89–103 †	104 **Rf**	105 **Db**	106 **Sg**	107 **Bh**	108 **Hs**	109 **Mt**	110 **Ds**	111 **Rg**	112 **Cn**	113 **Nh**	114 **Fl**	115 **Mc**	116 **Lv**	117 **Ts**	118 **Og**

*Lanthanoids	57 **La** 138.91	58 **Ce** 140.12	59 **Pr** 140.91	60 **Nd** 144.24	61 **Pm**	62 **Sm** 150.36	63 **Eu** 151.97	64 **Gd** 157.25	65 **Tb** 158.93	66 **Dy** 162.50	67 **Ho** 164.93	68 **Er** 167.26	69 **Tm** 168.93	70 **Yb** 173.05	71 **Lu** 174.97
†Actinoids	89 **Ac**	90 **Th** 232.04	91 **Pa** 231.04	92 **U** 238.03	93 **Np**	94 **Pu**	95 **Am**	96 **Cm**	97 **Bk**	98 **Cf**	99 **Es**	100 **Fm**	101 **Md**	102 **No**	103 **Lr**

AP® CHEMISTRY EQUATIONS AND CONSTANTS

UNIT SYMBOLS	
gram,	g
mole,	mol
liter,	L
meter,	m
second,	s
hertz,	Hz
atmosphere,	atm
millimeter of mercury,	mm Hg
degree Celsius,	°C
kelvin,	K
joule,	J
volt,	V
coulomb,	C
ampere,	A

UNIT CONVERSIONS
1 hertz = 1 s^{-1}
1 atm = 760 mm Hg = 760 torr
K = °C + 273.15
1 volt = $\frac{1 \text{ joule}}{1 \text{ coulomb}}$
1 ampere = $\frac{1 \text{ coulomb}}{1 \text{ second}}$

METRIC PREFIXES		
Factor	Prefix	Symbol
10^{9}	giga	G
10^{6}	mega	M
10^{3}	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

ATOMIC STRUCTURE

$$E = h\nu$$

$$c = \lambda\nu$$

$$F_{coulombic} \propto \frac{q_1 q_2}{r^2}$$

E = energy
ν = frequency
λ = wavelength
F = force
q = charge
r = separation
Planck's constant, $h = 6.626 \times 10^{-34}$ J s
Speed of light, $c = 2.998 \times 10^{8}$ m s^{-1}
Avogadro's number = 6.022×10^{23} mol^{-1}

GASES, LIQUIDS, AND SOLUTIONS

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$PV = nRT$$

$$P_A = P_{\text{total}} \times X_A, \text{ where } X_A = \frac{\text{moles A}}{\text{total moles}}$$

$$P_{\text{total}} = P_A + P_B + P_C + \ldots$$

$$n = \frac{m}{\boldsymbol{M}}$$

$$D = \frac{m}{V}$$

$$KE = \frac{1}{2}mv^2$$

$$M = \frac{n_{solute}}{\text{L}_{solution}}$$

$$A = \varepsilon bc$$

P = pressure
V = volume
T = temperature
n = number of moles
X = mole fraction
m = mass
$\boldsymbol{M}$ = molar mass
D = density
KE = kinetic energy
v = velocity
M = molarity
A = absorbance
ε = molar absorptivity
b = path length
c = concentration
Gas constant, $R = 8.314$ J mol^{-1} K^{-1}
$= 0.08206$ L atm K^{-1} mol^{-1}
STP = 273.15 K and 1.0 atm
Ideal gas at STP = 22.4 L mol^{-1}

KINETICS

$$[\mathrm{A}]_t - [\mathrm{A}]_0 = -kt$$

$$\ln[\mathrm{A}]_t - \ln[\mathrm{A}]_0 = -kt$$

$$\frac{1}{[\mathrm{A}]_t} - \frac{1}{[\mathrm{A}]_0} = kt$$

$$t_{1/2} = \frac{0.693}{k}$$

k = rate constant
t = time
$t_{1/2}$ = half-life

EQUILIBRIUM

$$K_c = \frac{[\mathrm{C}]^c[\mathrm{D}]^d}{[\mathrm{A}]^a[\mathrm{B}]^b}, \text{ where } a\ \mathrm{A} + b\ \mathrm{B} \rightleftarrows c\ \mathrm{C} + d\ \mathrm{D}$$

$$K_p = \frac{(P_\mathrm{C})^c(P_\mathrm{D})^d}{(P_\mathrm{A})^a(P_\mathrm{B})^b}$$

$$K_w = [\mathrm{H_3O^+}][\mathrm{OH^-}] = 1.0 \times 10^{-14} \text{ at } 25°\mathrm{C}$$

$$\mathrm{p}K_w = 14 = \mathrm{pH} + \mathrm{pOH} \text{ at } 25°\mathrm{C}$$

$$\mathrm{pH} = -\log[\mathrm{H_3O^+}], \qquad \mathrm{pOH} = -\log[\mathrm{OH^-}]$$

$$K_a = \frac{[\mathrm{H_3O^+}][\mathrm{A^-}]}{[\mathrm{HA}]}, \qquad K_b = \frac{[\mathrm{OH^-}][\mathrm{HB^+}]}{[\mathrm{B}]}$$

$$\mathrm{p}K_a = -\log K_a, \qquad \mathrm{p}K_b = -\log K_b$$

$$K_w = K_a \times K_b, \qquad \mathrm{p}K_w = \mathrm{p}K_a + \mathrm{p}K_b$$

$$\mathrm{pH} = \mathrm{p}K_a + \log\frac{[\mathrm{A^-}]}{[\mathrm{HA}]}$$

Equilibrium Constants

K_c (molar concentrations)
K_p (gas pressures)
K_w (water)
K_a (acid)
K_b (base)

THERMODYNAMICS/ELECTROCHEMISTRY

$$q = mc\Delta T$$

$$\Delta H^\circ_{reaction} = \sum \Delta H^\circ_{f\ products} - \sum \Delta H^\circ_{f\ reactants}$$

$$\Delta S^\circ_{reaction} = \sum S^\circ_{products} - \sum S^\circ_{reactants}$$

$$\Delta G^\circ_{reaction} = \sum \Delta G^\circ_{f\ products} - \sum \Delta G^\circ_{f\ reactants}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$
$$= -RT\ln K$$
$$= -nFE^\circ$$

$$I = \frac{q}{t}$$

$$E_{cell} = E^\circ_{cell} - \frac{RT}{nF}\ln Q$$

q = heat
m = mass
c = specific heat capacity
T = temperature
$S°$ = standard entropy
$H°$ = standard enthalpy
$G°$ = standard Gibbs free energy
R = gas constant
K = equilibrium constant
n = number of moles of electrons
$E°$ = standard potential
I = current (amperes)
q = charge (coulombs)
t = time (seconds)
Q = reaction quotient
Faraday's constant, F = 96,485 coulombs / 1 mol e^-

AP® CHEMISTRY

Section I

Time—1 hour and 30 minutes

60 Questions

CALCULATORS ARE ALLOWED FOR SECTION I.

Note: For all questions, assume that the temperature is 298 K, the pressure is 1.0 atm, and solutions are aqueous unless otherwise specified.

Directions: Each of the questions or incomplete statements below is followed by four suggested answers or completions. Select the one that is best in each case and then enter the letter in the corresponding space on the answer sheet.

1. Which of the following bonds is most polar?

(A) F–F

(B) O–F

(C) O–I

(D) O–Cl

2. A sample of an oxide of vanadium is separated into its elements, resulting in 1.02 g V and 0.48 g O. Which of the following is the formula of the oxide?

(A) V_2O

(B) VO_2

(C) V_2O_3

(D) V_2O_5

$$H_2O(g) \rightarrow H_2O(l)$$

3. Drops of water are observed to condense on a cold window, as represented by the equation. Which of the following is true about the values of $\Delta H°$ and $\Delta S°$ for the process?

(A) $\Delta H°$ and $\Delta S°$ are positive.

(B) $\Delta H°$ is positive, and $\Delta S°$ is negative.

(C) $\Delta H°$ is negative, and $\Delta S°$ is positive.

(D) $\Delta H°$ and $\Delta S°$ are negative.

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$$AlCl_3(aq) + 3\ NaOH(aq) \rightarrow Al(OH)_3(s) + 3\ NaCl(aq)$$

4. Which of the following is the balanced net ionic equation for the reaction represented by the preceding equation?

(A) $Al^{3+}(aq) + OH^-(aq) \rightarrow Al(OH)_3(s)$

(B) $Al^{3+}(aq) + 3\ OH^-(aq) \rightarrow Al(OH)_3(s)$

(C) $Al^{3+}(aq) + 3\ Cl^-(aq) + 3\ NaOH(aq) \rightarrow Al(OH)_3(s) + 3\ Na^+(aq) + 3\ Cl^-(aq)$

(D) $Al^{3+}(aq) + 3\ Cl^-(aq) + 3\ Na^+(aq) + 3\ OH^-(aq) \rightarrow Al(OH)_3(s) + 3\ Na^+(aq) + 3\ Cl^-(aq)$

$$Mn^{2+} + C_2O_4^{2-} \rightleftarrows MnC_2O_4 \qquad K_1 = 1 \times 10^3$$

$$[Mn(C_2O_4)_2]^{2-} \rightleftarrows MnC_2O_4 + C_2O_4^{2-} \qquad K_2 = 1 \times 10^{-2}$$

5. Which of the following is the value of the equilibrium constant for the following reaction, given the equilibrium constants K_1 and K_2?

$$Mn^{2+} + 2\ C_2O_4^{2-} \rightleftarrows [Mn(C_2O_4)_2]^{2-}$$

(A) 1×10^{-5}

(B) 1×10^{-1}

(C) 1×10^{0}

(D) 1×10^{5}

Reduction Half-Reaction	$E°$ at 298 K
$Pb^{2+} + 2\ e^- \rightarrow Pb$	–0.13 V
$Cr^{3+} + 3\ e^- \rightarrow Cr$	–0.74 V

6. Based on the table of half-reactions, which of the following is a true statement with correct justification?

(A) A reaction will occur when $Pb(s)$ is added to a 1.0 M solution of $Cr(NO_3)_3$, because $E°$ for the reaction is 0.61 V.

(B) A reaction will occur when $Pb(s)$ is added to a 1.0 M solution of $Cr(NO_3)_3$, because $E°$ for the reaction is 1.09 V.

(C) A reaction will occur when $Cr(s)$ is added to a 1.0 M solution of $Pb(NO_3)_2$, because $E°$ for the reaction is 0.61 V.

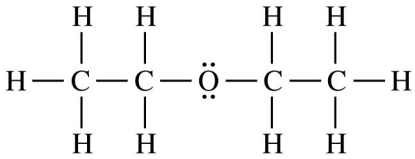
(D) A reaction will occur when $Cr(s)$ is added to a 1.0 M solution of $Pb(NO_3)_2$, because $E°$ for the reaction is 1.09 V.

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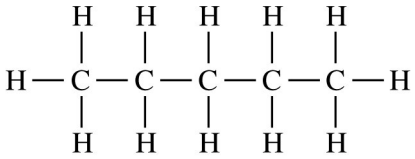
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7. Which of the following Lewis diagrams represents the substance that would exhibit the most significant hydrogen bonding in a pure liquid sample?

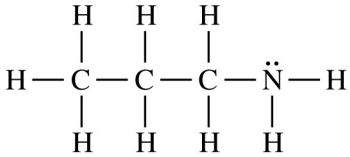
(A)



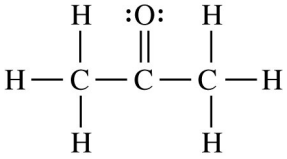
(B)



(C)



(D)



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Substance	Absolute Entropy $J/(mol \cdot K)$
$H_2(g)$	130
$Cl_2(g)$	220

$$H_2(g) + Cl_2(g) \rightarrow 2\ HCl(g) \quad \Delta S°_{rxn} = 20.\ J/(mol \cdot K)$$

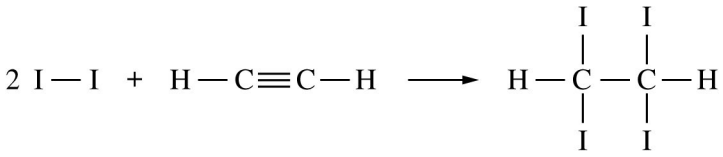
8. According to the information in the table and the chemical equation, the value of the molar entropy of $HCl(g)$ is closest to which of the following?

(A) $170\ J/(mol \cdot K)$

(B) $190\ J/(mol \cdot K)$

(C) $350\ J/(mol \cdot K)$

(D) $370\ J/(mol \cdot K)$



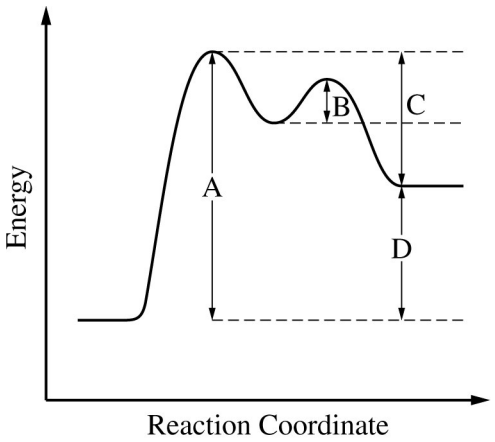
9. How does the hybridization of the carbon atoms change from reactants to products in the reaction represented by the equation?

(A) *sp* to sp^2

(B) *sp* to sp^3

(C) sp^2 to *sp*

(D) sp^3 to *sp*



10. Given the reaction energy profile, which of the following corresponds to the activation energy for the rate-determining step in the reaction mechanism?

(A) A

(B) B

(C) C

(D) D

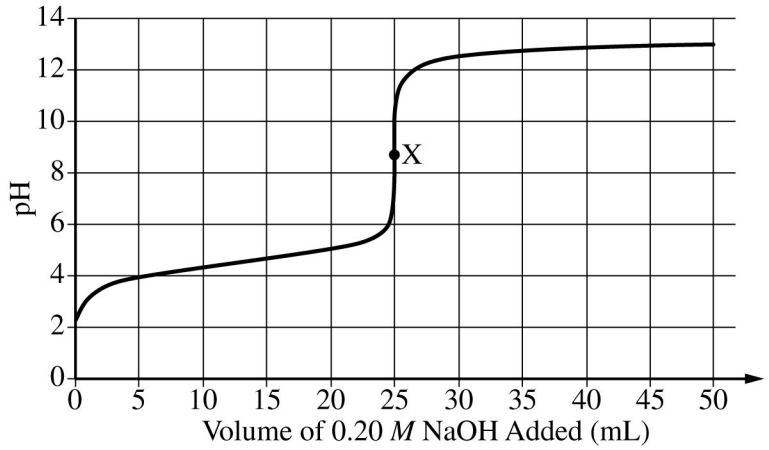
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Questions 11–14 refer to the following information.

$$HA(aq) + OH^-(aq) \rightarrow H_2O(l) + A^-(aq)$$

A 10.0 mL solution of an unknown acid, HA, is titrated with 0.20 *M* NaOH(aq). The reaction represented by the preceding equation occurs during the titration. The following titration curve is produced.



11. Which of the following is a conjugate acid-base pair for the reaction that occurs during the titration?

(A) HA and OH^-

(B) HA and H_2O

(C) H_2O and A^-

(D) OH^- and H_2O

12. The volume of the HA solution that was titrated was 10.0 mL. What was the initial concentration of the HA solution?

(A) 0.080 *M*

(B) 0.20 *M*

(C) 0.50 *M*

(D) 1.0 *M*

13. The pK_a of HA is closest to which of the following?

(A) 2.2

(B) 4.3

(C) 5.4

(D) 8.7

14. Which of the following shows the relative concentrations of species in the solution represented by point X on the titration curve?

(A) $[H^+] = [OH^-] > [A^-]$

(B) $[H^+] > [OH^-] = [A^-]$

(C) $[OH^-] > [A^-] > [H^+]$

(D) $[A^-] > [OH^-] > [H^+]$

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15. An unknown element, X, forms a compound with barium with the formula of BaX_2. Which of the following could be the electron configuration of a neutral atom of element X?

(A) [Ne] $3s^2$

(B) [Ne] $3s^2 3p^2$

(C) [Ne] $3s^2 3p^5$

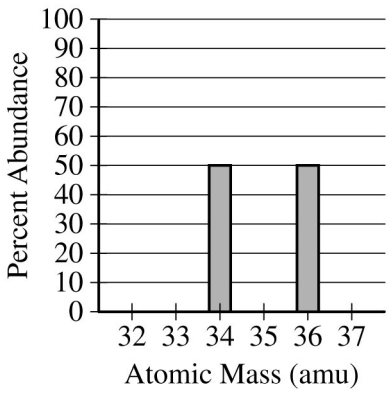
(D) [Ne] $3s^2 3p^6$

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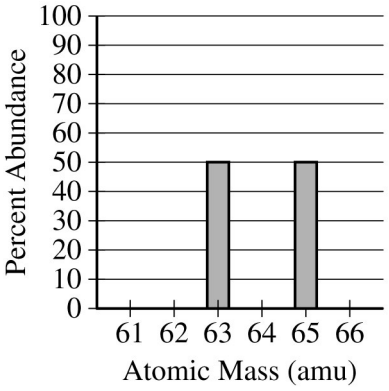
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16. Copper consists of two naturally occurring isotopes, one with 34 neutrons and one with 36 neutrons. Which of the following is the mass spectrum of a representative sample of Cu?

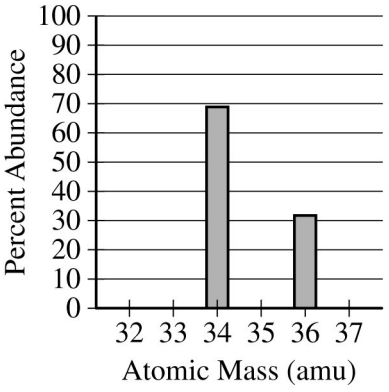
(A)



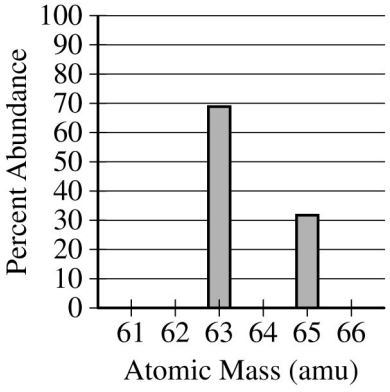
(B)



(C)



(D)



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Substance	Boiling Point (°C)	Molar Mass (g/mol)	K_a
CH_3OH	65	32.04	5.0×10^{-16}
H_2O	100	18.02	1.0×10^{-14}

17. Properties of CH_3OH and H_2O are given in the table. A student is given a beaker containing a solution of $CH_3OH(l)$ and $H_2O(l)$. If the student wishes to separate and collect the two components of the solution, which of the following experimental techniques would be most appropriate?

(A) Using paper chromatography and collecting the liquid that first moves to the top of the paper

(B) Combusting the CH_3OH and collecting the products of the reaction

(C) Boiling the mixture and condensing and collecting the first vapors produced

(D) Titrating the mixture with $HCl(aq)$ to the equivalence point

18. In a kinetic study of a reaction, a student measures concentration of one species in the reaction at regular time intervals and graphs the concentration versus time. If the reaction was zeroth order instead of first order, then the graph of concentration versus time would show a

(A) constant half-life

(B) constant slope

(C) half-life that gets progressively longer

(D) slope that becomes less steep over time

$$MgCO_3(s) \rightleftarrows MgO(s) + CO_2(g)$$
$$\Delta H^\circ = 100 \text{ kJ/mol}_{rxn}$$

19. A sample of $MgCO_3(s)$ is placed in a previously evacuated rigid container. The temperature is raised to 250°C, and the reaction represented by the equation reaches equilibrium with both solids present in the container. Which of the following adjustments to the system will increase the number of moles of $CO_2(g)$ at equilibrium?

(A) Raising the temperature of the system to 300°C

(B) Lowering the temperature of the system to 200°C

(C) Adding $MgO(s)$ to the system

(D) Adding $MgCO_3(s)$ to the system

$$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$$

20. A student conducts an experiment to determine the percentage of silver in an alloy by first reacting the alloy with nitric acid. The student then precipitates the $Ag^+(aq)$ by adding excess $NaCl(aq)$ to form $AgCl(s)$ according to the equation. The precipitate is filtered, dried, and weighed. Which of the following indicates the effect on the calculated percentage of silver if the precipitate was not dried completely before weighing it, and explains why?

(A) The calculated percentage is too low because the additional water will cause a significant quantity of the $AgCl(s)$ to dissolve and wash through the filter paper.

(B) The calculated percentage is too low because water has a significantly lower molar mass than $AgCl(s)$, which reduces the average mass of the precipitate.

(C) The calculated percentage is the same because the additional water does not affect the mass of the alloy, and the student will react the same mass of Ag regardless.

(D) The calculated percentage is too high because the additional water adds mass to the $AgCl(s)$, and the student will calculate a higher mass of Ag in the sample.

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Questions 21–24 refer to the following information.

An equimolar mixture of SO_2 and O_2 is placed in a rigid container at 298 K. The initial pressure in the container is 0.20 atm. A catalyst is added to the mixture, and the reaction represented below occurs.

$$2\ SO_2(g) + O_2(g) \rightarrow 2\ SO_3(g) \qquad \Delta H° = -200\ \text{kJ/mol}_{rxn}$$

21. Which of the following gives the shape of the SO_2 molecule and best helps explain why it has that shape?

(A) It is linear, because there are two electron domains around the sulfur atom.

(B) It is linear, because there are three electron domains around the sulfur atom.

(C) It is bent, because there are two electron domains around the sulfur atom.

(D) It is bent, because there are three electron domains around the sulfur atom.

22. What is the partial pressure of $O_2(g)$ in the container before any reaction occurs?

(A) 0.067 atm

(B) 0.10 atm

(C) 0.13 atm

(D) 0.20 atm

Bond Type	Bond Energy (kJ/mol)
Sulfur-oxygen bond in SO_2	530
Oxygen-oxygen bond in O_2	500
Sulfur-oxygen bond in SO_3	?

23. Based on the bond energies given in the table and the value of $\Delta H°$, which of the following is the best estimate for the energy of one of the sulfur-oxygen bonds in SO_3?

(A) 403 kJ/mol

(B) 470 kJ/mol

(C) 880 kJ/mol

(D) 1410 kJ/mol

24. What is the partial pressure of $SO_3(g)$ in the container after the reaction goes to completion and the temperature is returned to 298 K?

(A) 0.050 atm

(B) 0.10 atm

(C) 0.20 atm

(D) 0.40 atm

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Identity of sample	Sample X	Sample Z
Mass of ice placed in sample	6.0 g	6.0 g
Temperature change of sample	–5.0 K	–3.0 K

25. A student performs an experiment on samples of two unknown liquids. Each of the samples, which have the same mass, are placed into identical insulated cups. Identical masses of ice are added to each sample, and the ice melts completely. The student collects the information shown in the table. Which of the following could explain why the temperature changes in samples X and Z were different?

(A) Sample X and sample Z were of the same substance, but the volume of sample X was greater than that of sample Z.

(B) Sample X and sample Z were of the same substance, but the specific heat of sample X was less than that of sample Z.

(C) Sample X and sample Z were of different substances, but the amount of energy needed to melt the ice in sample X was greater than in sample Z.

(D) Sample X and sample Z were of different substances, and the specific heat of sample X was less than sample Z.

26. Humans are warned to shield their eyes from ultraviolet radiation with wavelengths shorter than 300 nm but not from infrared radiation with wavelengths of about 900 nm. Which of the following explains the reason for this based on what happens at the molecular level with exposure to these types of radiation?

(A) 900 nm radiation has triple the energy of 300 nm radiation, so infrared radiation is more likely to promote electrons or break covalent bonds.

(B) 900 nm radiation has one-third the energy of 300 nm radiation, so infrared radiation is more likely to cause molecular rotation to occur.

(C) 300 nm radiation has triple the energy of 900 nm radiation, so ultraviolet radiation is more likely to promote electrons or break covalent bonds.

(D) 300 nm radiation has one-third the energy of 900 nm radiation, so ultraviolet radiation is more likely to cause molecular vibration to occur.

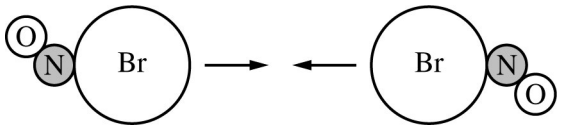
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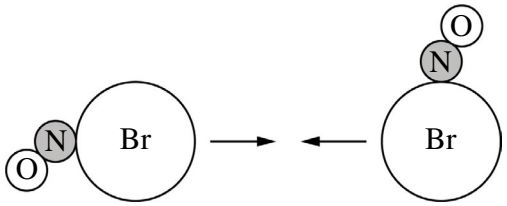
$$2\ \mathrm{NOBr} \rightarrow 2\ \mathrm{NO} + \mathrm{Br_2}$$

27. An elementary step in a reaction mechanism is given above. Which of the following diagrams would best help explain why not all collisions result in reaction?

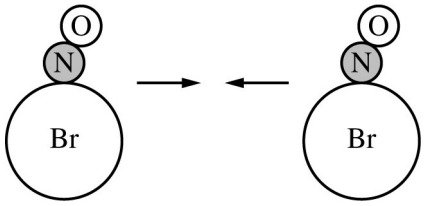
(A)



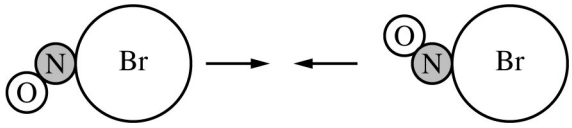
(B)



(C)



(D)



$$\mathrm{CO}(g) + \mathrm{NO_2}(g) \rightarrow \mathrm{CO_2}(g) + \mathrm{NO}(g)$$

28. The rate law for the reaction represented is rate = $k\left[\mathrm{NO_2}\right]^2$. Which of the following mechanisms is consistent with the rate law?

(A) $2\ \mathrm{NO_2} \rightarrow 2\ \mathrm{O_2} + \mathrm{N_2}$ *fast*
$\mathrm{N_2} + \mathrm{CO} \rightarrow \mathrm{N_2O} + \mathrm{C}$ *slow*
$\mathrm{C} + \mathrm{NO_2} \rightarrow \mathrm{CO_2} + \mathrm{N}$ *fast*
$2\ \mathrm{N} + \mathrm{O_2} \rightarrow 2\ \mathrm{NO}$ *fast*

(B) $2\ \mathrm{NO_2} \rightleftarrows \mathrm{N_2O_4}$ *fast*
$\mathrm{N_2O_4} + 2\ \mathrm{CO} \rightarrow 2\ \mathrm{CO_2} + 2\ \mathrm{NO}$ *slow*

(C) $2\ \mathrm{NO_2} \rightarrow 2\ \mathrm{NO} + \mathrm{O_2}$ *slow*
$\mathrm{O_2} + \mathrm{CO} \rightarrow \mathrm{CO_2} + \mathrm{O}$ *fast*
$\mathrm{O} + \mathrm{CO} \rightarrow \mathrm{CO_2}$ *fast*

(D) $2\ \mathrm{NO_2} \rightarrow \mathrm{NO} + \mathrm{NO_3}$ *fast*
$\mathrm{NO_3} + \mathrm{CO} \rightarrow \mathrm{NO_2} + \mathrm{CO_2}$ *slow*

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Questions 29–31 refer to the following information.

Element	Radius	Electronegativity
Ag	165 pm	1.93
Au	174 pm	2.54
Cu	145 pm	1.90
N	56 pm	3.04
Ni	149 pm	1.91
W	193 pm	2.36

29. Which of the following names a pair of elements that would likely form a substitutional alloy and explains why?

(A) Ni and Cu, because Ni atoms would fit in the spaces between Cu atoms in the solid lattice.

(B) Ni and Cu, because Ni atoms would take the place of Cu atoms in the solid lattice.

(C) W and N, because N atoms would fit in the spaces between W atoms in the solid lattice.

(D) W and N, because N atoms would take the place of W atoms in the solid lattice.

30. Which of the following explains why an alloy of Au and Ag conducts electricity?

(A) Ag typically forms only Ag^+ ions, whereas Au forms both Au^+ and Au^{3+} ions.

(B) Au has a larger atomic radius than Ag, so its electrons are easier to remove.

(C) Both Ag and Au have delocalized valence electrons.

(D) Both Ag and Au attract shared electrons equally.

31. Which of the following best explains why Au has a larger radius than Cu does?

(A) Au has a larger electronegativity than Cu does.

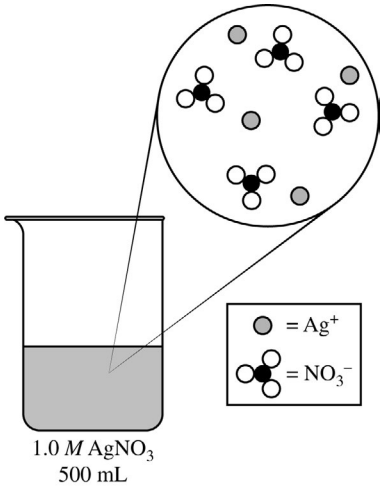
(B) Au has more occupied electron shells than Cu does.

(C) Au has more protons than Cu does.

(D) Au is more dense than Cu is.

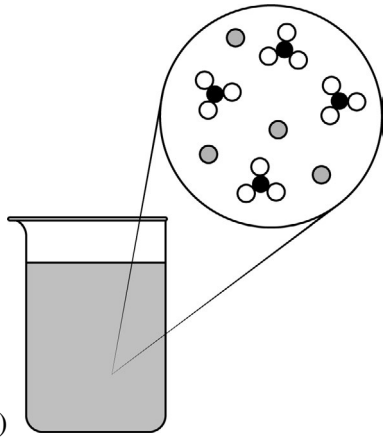
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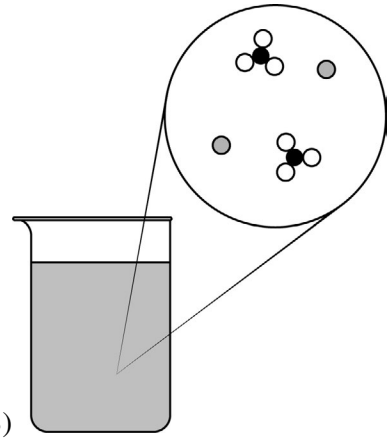


32. A 1.0 *M* solution of $AgNO_3$ is prepared by dissolving 0.50 mol of $AgNO_3(s)$ in water to make 500 mL of solution. A highly magnified view of the solution is shown. If 1.50 mol of $AgNO_3(s)$ is added to the solution in the beaker and diluted to a final volume of 1000 mL, which of the following magnified views most accurately reflects the composition of the solution? (All magnified views represent the same volume of solution.)

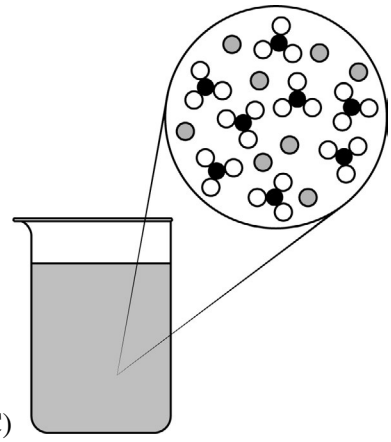
(A)



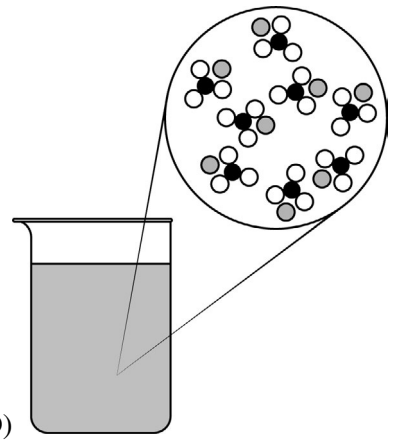
(B)



(C)



(D)



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33. A solution is prepared by mixing 0.50 L of 0.40 *M* HNO_3 with 0.50 L of 0.20 *M* KOH. If it is assumed that volumes are additive, the pH of the resulting mixture will be which of the following?

(A) 0.10

(B) 0.70

(C) 1.0

(D) 7.0

$$Mg(s) + 2\ HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

34. A student reacts $Mg(s)$ with $HCl(aq)$ to determine the value of $\Delta H°$ for the reaction represented. Information for the experiment is given in the following table.

Mass of $Mg(s)$ used	2.4 g
Volume of 1.0 *M* HCl used	300. mL
Temperature change of solution	15.1 K
Specific heat of solution	4.0 J/(g·K)

If the student conducts a second trial and uses the same mass of Mg(*s*) but 500. mL of 1.0 *M* HCl, which of the following will occur?

(A) The value of the calculated $\Delta H°$ will increase.

(B) The temperature change of the solution will be less than 15.1 K.

(C) The temperature change of the solution will be approximately 25 K.

(D) The temperature change of the solution will be 15.1 K.

Compound	Molar Mass	K_{sp}	Appearance
NaOH	40. g/mol	1.1×10^{2}	White, flaky powder
KOH	56 g/mol	3.6×10^{2}	White powder
$Mg(OH)_2$	58 g/mol	1.8×10^{-11}	White powder
$Ca(OH)_2$	74 g/mol	5.5×10^{-6}	White, flaky powder

35. Formulas and properties of some group 1 and 2 compounds are listed in the table. Which of the following questions could be tested to determine if an unknown white powder was a hydroxide from group 2 and not from group 1?

(A) Does the compound conduct electricity when dissolved in distilled water?

(B) Does the compound dissolve significantly in distilled water?

(C) Does the compound form a precipitate when reacted with HNO_3?

(D) Does the compound have a molar mass above 50 g/mol?

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Questions 36–38 refer to the following information.

$$2\ \mathrm{NaCl}(aq) + 2\ \mathrm{H_2O}(l) \rightarrow 2\ \mathrm{NaOH}(aq) + \mathrm{Cl_2}(g) + \mathrm{H_2}(g)\ \ \Delta G^\circ = 400\ \mathrm{kJ\ /\ mol}_{rxn}$$

Sodium hydroxide can be produced commercially from the electrolysis of sodium chloride solutions, as indicated by the balanced equation.

36. Which of the following is the best evidence that the reaction is an oxidation-reduction reaction?

(A) Heat is consumed during the reaction.

(B) The oxidation numbers of both Na and O change as the reaction proceeds.

(C) The oxidation number of each Cl atom changes from -1 to 0, and some H atoms change from $+1$ to 0.

(D) Each Cl atom gains an electron, and each $\mathrm{H_2O}$ molecule loses a proton.

37. What is the approximate minimum voltage that must be supplied to a 1.0 M solution of $\mathrm{NaCl}(aq)$ to produce $\mathrm{NaOH}(aq)$ according to the reaction above? ($F \approx 100{,}000$ coulombs per mole of electrons.)

(A) 0.004 V

(B) 1.0 V

(C) 2.0 V

(D) 4.0 V

38. The gases produced from the electrolysis reaction are condensed in order to prevent their escape into the atmosphere. Which of the following indicates the gas that condenses at the lower temperature and gives a particle-level explanation for why?

(A) $\mathrm{H_2}$, because the hydrogen bonding in $\mathrm{H_2}(l)$ is stronger than the dispersion forces in $\mathrm{Cl_2}(l)$.

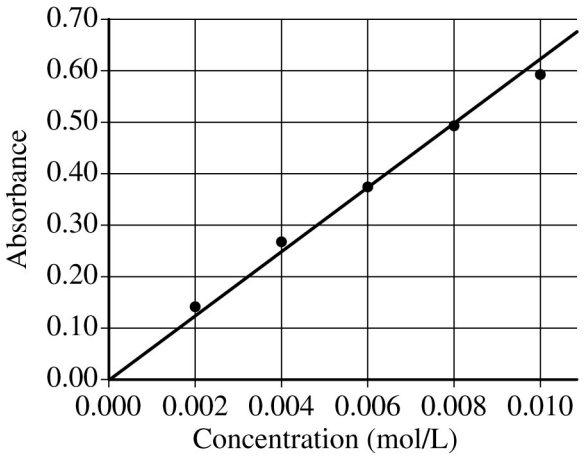
(B) $\mathrm{H_2}$, because the dispersion forces in $\mathrm{H_2}(l)$ are weaker than the dispersion forces in $\mathrm{Cl_2}(l)$.

(C) $\mathrm{Cl_2}$, because the $\mathrm{Cl-Cl}$ bond in $\mathrm{Cl_2}$ is stronger than the $\mathrm{H-H}$ bond in $\mathrm{H_2}$.

(D) $\mathrm{Cl_2}$, because Cl atoms are more electronegative than H atoms.

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39. A student prepares several solutions of known concentration and measures their absorbance in a 1.0 cm cuvette to generate the Beer-Lambert plot shown. Another student uses the same solutions with a 2.0 cm cuvette. Which of the following indicates how the second student's plot will differ from the plot of the first student?

(A) The slope will be doubled because the absorbance is doubled in the 2.0 cm cell.

(B) The slope will be halved because half as much light gets through the 2.0 cm cell.

(C) The slope will be doubled because the molar absorptivity is doubled in the 2.0 cm cuvette.

(D) The slope will be halved because the molar absorptivity is halved in the 2.0 cm cuvette.

40. A student measures the ratio $\frac{PV}{nT}$ for a sample of gaseous C_4H_{10} at three different volumes in a sealed syringe of variable volume. At all times the syringe is held at a temperature of 300 K. The data are shown in the table.

Volume	Ratio $\frac{PV}{nT}$
20.0 mL	0.0821
10.0 mL	0.0790
5.00 mL	0.0718

Which of the following best explains the trend in the data as the volume decreases?

(A) The C_4H_{10} molecules decompose as the volume decreases, creating more moles of gas.

(B) The volume of the C_4H_{10} molecules becomes more significant as the volume decreases.

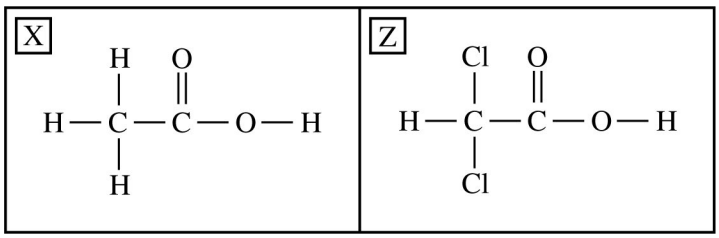
(C) The intermolecular forces between C_4H_{10} molecules become more significant as the volume decreases.

(D) The average kinetic energy of the C_4H_{10} molecules decreases as the volume decreases.

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41. Structural diagrams for two weak acids, X and Z, are shown.



Which acid has the larger value of K_a, and why?

(A) X, because more hydrogen atoms per molecule can dissociate, decreasing the pH of the solution.

(B) X, because the weaker London dispersion forces make the hydrogen atom more likely to be separated from the conjugate base.

(C) Z, because the symmetry of the molecule results in a weaker dipole and a less polarizable O – H bond, making the hydrogen atom less likely to be separated from the conjugate base.

(D) Z, because the electronegative Cl atoms stabilize the conjugate base, allowing the acid to ionize to a greater degree than acid X.

$$2\ H_2O_2 \rightarrow 2\ H_2O + O_2$$

42. The reaction represented is first order with respect to H_2O_2. An experiment is performed in which 100. mL of 0.20 M H_2O_2 is decomposed to H_2O and O_2 at a temperature of 300 K. A chemist wants to triple the rate of the reaction. Which of the following experimental conditions would triple the rate of the reaction?

(A) Repeating the experiment with 50. mL of 0.60 M H_2O_2 at 300 K

(B) Repeating the experiment with 100. mL of 0.20 M H_2O_2 at 100 K

(C) Repeating the experiment with 200. mL of 0.067 M H_2O_2 at 300 K

(D) Repeating the experiment with 300. mL of 0.20 M H_2O_2 at 300 K

Questions 43–45 refer to the following information.

$$N_2(g) + 3\ H_2(g) \rightarrow 2\ NH_3(g)$$

Samples of $N_2(g)$ and $H_2(g)$ are placed in a rigid container at a constant temperature of 315 K and allowed to react according to the equation shown.

43. Which of the following compares the average kinetic energy and average speed of the $N_2(g)$ and $H_2(g)$ molecules in the container at 315 K before any reaction occurs?

(A) The $H_2(g)$ molecules have a greater average kinetic energy and average speed than the $N_2(g)$ molecules.

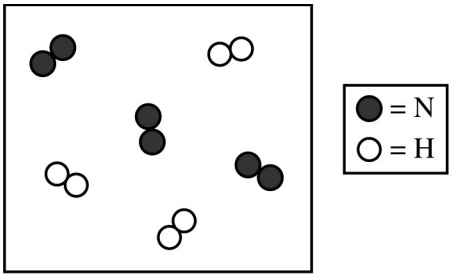
(B) The $H_2(g)$ molecules have a greater average kinetic energy than the $N_2(g)$ molecules, but the molecules have the same average speed.

(C) The $H_2(g)$ molecules have a greater average speed than the $N_2(g)$ molecules, but the molecules have the same average kinetic energy.

(D) The $H_2(g)$ molecules and $N_2(g)$ molecules have the same average kinetic energy and average speed.

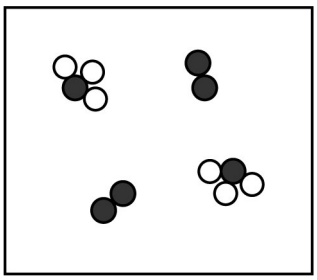
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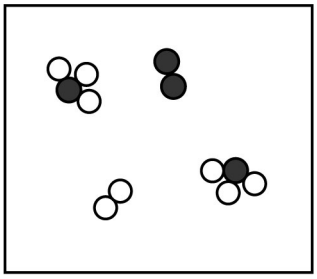


44. The reactants are represented in the diagram. A catalyst is added to the mixture and the reaction occurs. Which of the following is a correct particulate representation of the mixture after the reaction occurs?

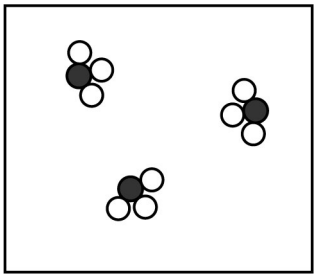
(A)



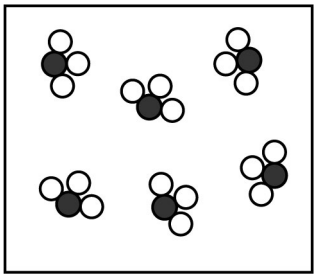
(B)



(C)



(D)



45. The synthesis of ammonia is carried out in three glass vessels of equal volume. In each vessel, N_2 and H_2 both have an initial partial pressure of 3.0 atm. A small piece of iron is added to one vessel, and a small piece of nickel is added to another vessel. The temperature is kept constant, and the pressure is measured at regular intervals, as shown in the table.

	Total Pressure in Reaction Vessel		
Time	Glass	Glass with Iron Added	Glass with Nickel Added
0 min	6.0 atm	6.0 atm	6.0 atm
30 min	6.0 atm	5.8 atm	5.8 atm
60 min	6.0 atm	5.6 atm	5.6 atm
90 min	6.0 atm	5.5 atm	5.5 atm

Which of the following explains the results shown in the table?

(A) Both iron and nickel are inhibitors; they prevent the system from reaching equilibrium.

(B) Both iron and nickel are catalysts; more N_2 and H_2 are consumed in the iron and nickel vessels than in the glass vessel, reducing the overall pressure.

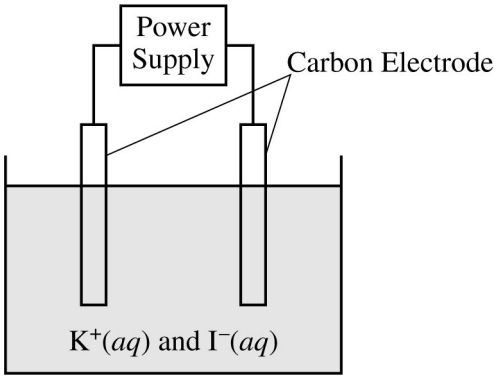
(C) Both iron and nickel react strongly with NH_3, causing it to decompose into N_2 and H_2, reducing the overall pressure.

(D) Solids do not appear in the equilibrium expression, so the partial pressures of the gases are not affected by the metal.

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46. A solution of KI(aq) is electrolyzed using the setup shown.



Bubbles appear on the cathode, and a yellow color appears in the solution surrounding the anode. When phenolphthalein ($pK_a = 9.4$) is added, the solution surrounding the cathode turns pink. Standard reduction potentials are given in the table.

Half-Reaction	$E°$(V)
$K^+(aq) + e^- \rightarrow K(s)$	–2.92 V
$H_2O(l) + 2\ e^- \rightarrow H_2(g) + 2\ OH^-(aq)$	–0.83 V
$O_2(g) + 2\ H_2O(l) + 4\ e^- \rightarrow 4\ OH^-(aq)$	0.40 V
$I_2(aq) + 2\ e^- \rightarrow 2\ I^-(aq)$	0.54 V
$O_2(g) + 4\ H^+(aq) + 4\ e^- \rightarrow 2\ H_2O(l)$	1.23 V

Which of the following identifies what is produced at the cathode and provides correct reasoning?

(A) $K(s)$ is produced; the phenolphthalein indicates that the solution is acidic, and alkali metals produce $H_2(g)$ and acidic solutions when added to water.

(B) $H_2(g)$ and $OH^-(aq)$ are produced; gas bubbles form, and the phenolphthalein color change indicates that $OH^-(aq)$ is produced in the solution around the cathode.

(C) $O_2(g)$ and $OH^-(aq)$ are produced; gas bubbles form, and the phenolphthalein color change indicates that $OH^-(aq)$ is produced in the solution around the cathode.

(D) $O_2(g)$ is produced; the last half-reaction in the table has the most positive reduction potential, which indicates that it is the most favored reaction.

47. A student conducts an experiment to measure the specific heat capacity, c, of a metal. The student heats 20.0 g of the metal, adds the heated metal into a calorimeter containing 100.0 g of water initially at 18.5°C, and the mixture comes to thermal equilibrium at 19.7°C. Which of the following experimental errors best explains why the value of c calculated from the data is smaller than the accepted value for the metal?

(A) The heated metal transfers some thermal energy to the surrounding air before it contacts the water in the calorimeter.

(B) The warmer air in the laboratory transfers thermal energy into the colder water in the calorimeter.

(C) An improperly calibrated balance results in more than 20.0 g of metal being used in the experiment.

(D) An improperly calibrated balance results in less than 100.0 g of water being used in the experiment.

$$2\,N_2(g) + O_2(g) \rightleftarrows 2\,N_2O(g) \qquad K_{eq} = K_1$$

48. The reaction represented has an equilibrium constant with a value of K_1. What is the value of the equilibrium constant, in terms of K_1, for the reaction $4\,N_2O(g) \rightleftarrows 4\,N_2(g) + 2\,O_2(g)$?

(A) $(K_1)^2$

(B) $\dfrac{1}{\sqrt{K_1}}$

(C) $\dfrac{1}{2K_1}$

(D) $\dfrac{1}{(K_1)^2}$

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$$CO(g) + H_2O(g) \rightleftarrows CO_2(g) + H_2(g) \quad K_p = 100 \text{ at } 500 \text{ K}$$

Substance	Initial Partial Pressure (atm)
$CO(g)$	0.50
$H_2O(g)$	0.10
$CO_2(g)$	6.0
$H_2(g)$	1.0

49. A mixture of $CO(g)$, $H_2O(g)$, $CO_2(g)$, and $H_2(g)$ is placed in a previously evacuated, rigid container. The initial partial pressures are given in the table. Which of the following predicts what will happen to the concentration of CO_2 as the system approaches equilibrium at 500 K, with justification?

(A) $[CO_2]$ will increase because $Q > K$.

(B) $[CO_2]$ will increase because $Q < K$.

(C) $[CO_2]$ will decrease because $Q > K$.

(D) $[CO_2]$ will decrease because $Q < K$.

50. Which of the following reactions has the most negative value of $\Delta S°$?

(A) $HCl(aq) + NaOH(aq) \rightarrow H_2O(l) + NaCl(aq)$

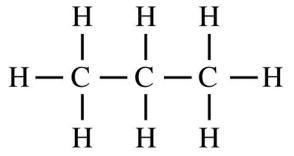
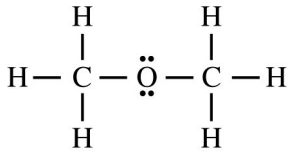
(B) $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

(C) $C_2H_2(g) + 2\ Cl_2(g) \rightarrow C_2H_2Cl_4(l)$

(D) $HCN(g) \rightarrow HNC(g)$

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Compound	Propane	Dimethyl ether
Lewis diagram		
Solubility in 100 g of H_2O	0.007 g	7 g

51. Information about two compounds is given in the table. Which of the following provides particle-level reasoning for the solubility of the two compounds?

(A) Molecules of both compounds are polar and will experience London dispersion forces only.

(B) Molecules of both compounds have similar masses and therefore have similar London dispersion forces.

(C) Molecules of propane have more available hydrogen bonding sites for H_2O molecules.

(D) Molecules of dimethyl ether will experience hydrogen bonding attractions to H_2O molecules, but molecules of propane will not.

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Questions 52–55 refer to the following information.

A solution is prepared by combining 1.0 L of 0.50 *M* $HClO_2$ with 1.0 L of 0.50 *M* $KClO_2$. The pH of the solution is 2.0.

52. The value of K_b for ClO_2^- is closest to which of the following?

(A) 1.0×10^{-14}

(B) 1.0×10^{-12}

(C) 1.0×10^{-7}

(D) 1.0×10^{-2}

53. When 10 mL of 0.50 *M* NaOH is added to the solution, the pH does not change significantly. Which of the following is the net ionic equation for the reaction that occurs and prevents a significant change in pH?

(A) $NaOH(aq) + K^+(aq) \rightarrow KOH(aq) + Na^+(aq)$

(B) $NaOH(aq) + ClO_2^-(aq) \rightarrow NaCl(aq) + H_2O(l)$

(C) $OH^-(aq) + Na^+(aq) \rightarrow NaOH(s)$

(D) $OH^-(aq) + HClO_2(aq) \rightarrow ClO_2^-(aq) + H_2O(l)$

54. If the solution was prepared by combining 1.0 L of 2.0 *M* $HClO_2$ and 1.0 L of 2.0 *M* $KClO_2$ instead of 1.0 L of 0.50 *M* $HClO_2$ and 1.0 L of 0.50 *M* $KClO_2$, which of the following would describe the solution?

(A) It would have a pH greater than 2.0.

(B) It would have a pH less than 2.0.

(C) It would have a pH of 2.0 and a greater buffer capacity than the solution made with 0.50 *M* $HClO_2$ and $KClO_2$.

(D) It would have a pH of 2.0 and a lesser buffer capacity than the solution made with 0.50 *M* $HClO_2$ and $KClO_2$.

55. Consider the Lewis electron-dot diagram for the ClO_2^- ion that minimizes formal charge. Which of the following is true about the hybridization of the Cl atom in the ion?

(A) The Cl atom is not hybridized, and the ion contains one single Cl – O bond and one double Cl – O bond.

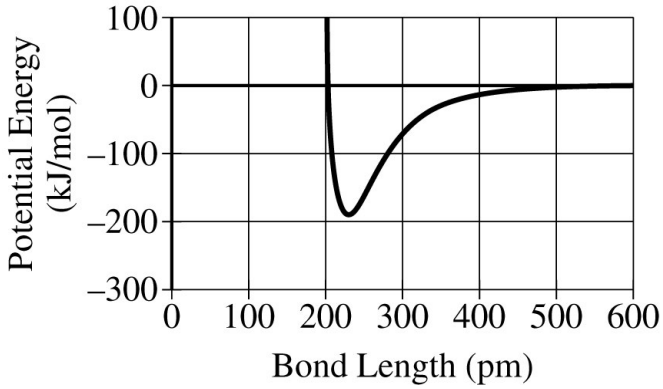
(B) The Cl atom is *sp* hybridized, and the ion contains two equivalent Cl – O double bonds.

(C) The Cl atom is sp^2 hybridized, and the ion contains two equivalent Cl – O single bonds.

(D) The Cl atom is sp^3 hybridized, and the ion contains two equivalent Cl – O single bonds.

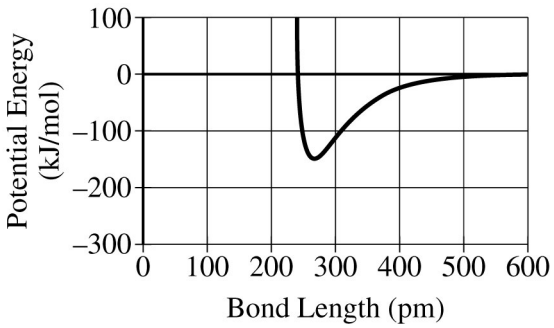
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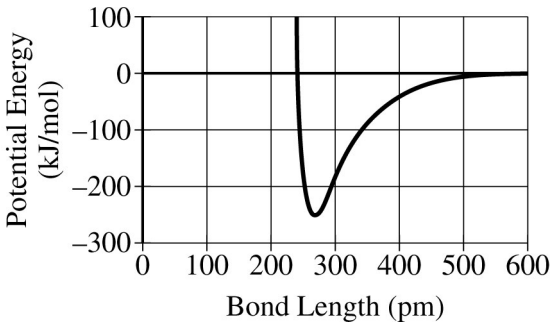


56. The preceding graph shows potential energy versus bond length for a Br_2 molecule. Which of the following graphs shows potential energy versus bond length for an I_2 molecule?

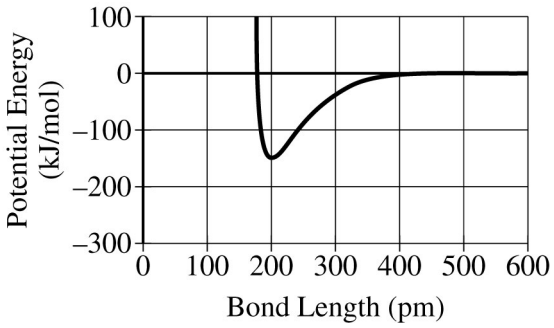
(A)



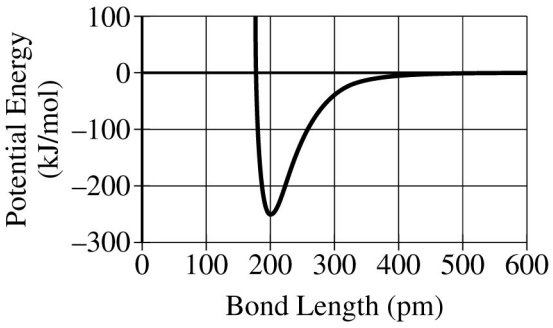
(B)



(C)

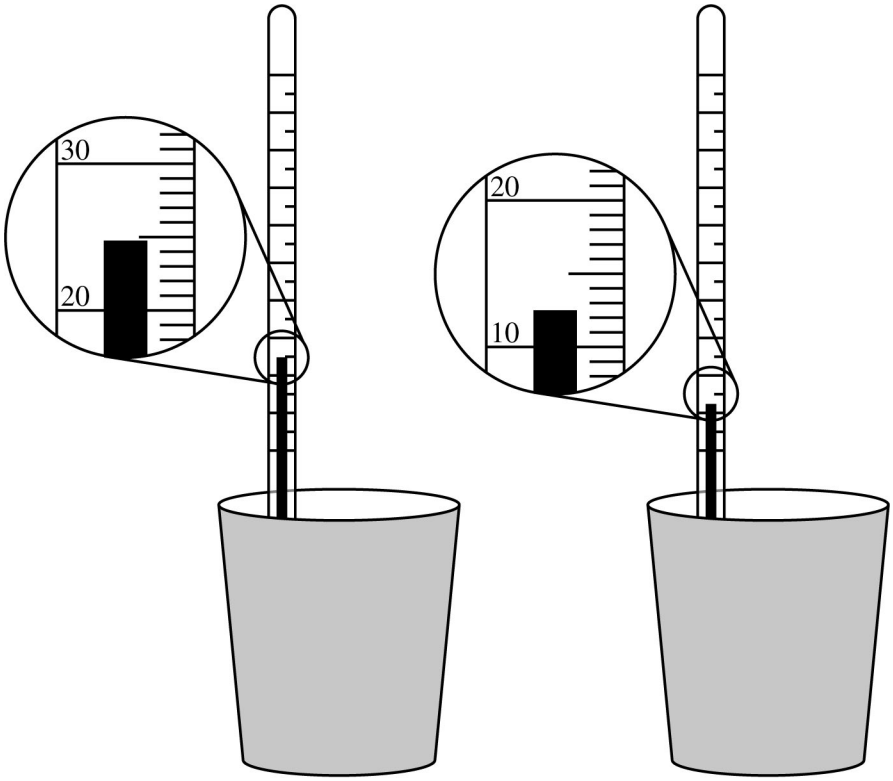


(D)



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Before NH_4Cl Added After NH_4Cl Added

57. To measure the heat of solution for NH_4Cl, a student measures the temperature of 100.0 g of H_2O (specific heat 4.18 J/(g·°C)), adds 22.15 g of NH_4Cl, and measures the temperature after the NH_4Cl is dissolved. The temperatures are shown in the diagram. Which of the following indicates the sign of ΔH_{soln} and the number of significant figures that can be reported for the value of ΔH_{soln} in the experiment? (Assume that the specific heat of the solution is identical to the specific heat of water.)

(A) $\Delta H_{soln} < 0$, 2 significant figures

(B) $\Delta H_{soln} < 0$, 3 significant figures

(C) $\Delta H_{soln} > 0$, 2 significant figures

(D) $\Delta H_{soln} > 0$, 3 significant figures

58. The molar solubility of PbI_2 is 1×10^{-3} M. Which of the following is the best estimate for the value of K_{sp} of PbI_2?

(A) 1×10^{-6}

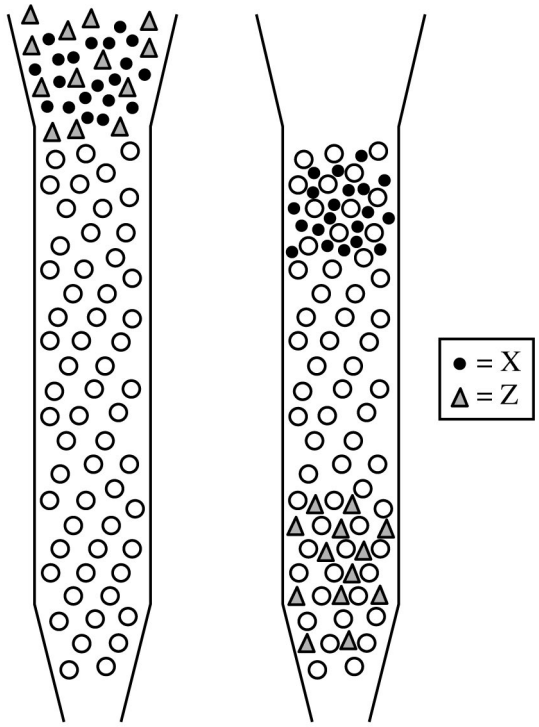
(B) 2×10^{-6}

(C) 1×10^{-9}

(D) 4×10^{-9}

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59. A chemist conducts a column chromatography experiment by placing an aqueous mixture of X and Z into a column that is packed with a nonpolar stationary phase, as represented on the left. After ethanol $\left(C_2H_5OH\right)$ is poured through the column, separation of X and Z occurs, as represented by the column on the right. Which of the following explains the experimental result based on particle interactions?

(A) X is more polar than Z; molecules of X are more attracted to the stationary phase than to the mobile phase.

(B) X has a smaller molecular volume than Z does; molecules of X pass more easily through the stationary phase.

(C) Z is more polar than X; molecules of Z are more attracted to the mobile phase than to the stationary phase.

(D) Z has a larger molar mass than X does; molecules of Z are more influenced by gravity.

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Reaction 1: $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$ $\Delta H^\circ = -285$ kJ/mol$_{rxn}$

Reaction 2: $H_2(g) + O_2(g) \rightarrow H_2O_2(aq)$ $\Delta H^\circ = ?$

Reaction 3: $2\ H_2O_2(aq) \rightarrow 2\ H_2O(l) + O_2(g)$ $\Delta H^\circ = -200.$ kJ/mol$_{rxn}$

60. Reactions 1 and 2 represented can be manipulated to produce reaction 3. What is the value of ΔH° for reaction 2?

(A) −370 kJ/mol$_{rxn}$

(B) −185 kJ/mol$_{rxn}$

(C) 85 kJ/mol$_{rxn}$

(D) 185 kJ/mol$_{rxn}$

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END OF SECTION I
IF YOU FINISH BEFORE TIME IS CALLED,
YOU MAY CHECK YOUR WORK ON THIS SECTION.
DO NOT GO ON TO SECTION II UNTIL YOU ARE TOLD TO DO SO.

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